THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 5–8: **Measurement of Length and Motion, Materials Around Us, Temperature and its Measurement, A Journey through States of Water** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The length of a table is 1.2 m. In centimetres it is: 1

(a) 1200 cm **(b)** 12 cm **(c)** 0.12 cm **(d)** 120 cm

2. A student places a stone, a cork, an iron nail and a leaf in water. Which pair floats? 1

(a) cork and leaf **(b)** nail and stone **(c)** stone and cork **(d)** leaf and nail

3. A clinical thermometer shows 38.5 °C. The person has: 1

(a) normal temperature **(b)** no temperature

(c) fever **(d)** low temperature

4. A glass of water left in an open window for a week shows less water. This is because of: 1

(a) freezing **(b)** evaporation **(c)** melting **(d)** condensation

5. Which motion is both periodic and linear (back and forth along a line)? 1

(a) a wheel **(b)** a falling stone

(c) a spinning top **(d)** a mass vibrating on a spring

6. Why does a stone sink in water but a boat made of iron float? 1

(a) iron is lighter than stone **(b)** boats have no weight

(c) the boat's shape holds a lot of air **(d)** water pushes iron

7. Why does the reading of a clinical thermometer stay after it is removed from the mouth? 1

(a) because of the bulb **(b)** because of the scale

(c) because of the kink **(d)** because of the glass

8. Cloud formation requires: 1

(a) freezing and melting **(b)** boiling only

(c) evaporation and condensation **(d)** melting and freezing

9. The metal hook at the end of a measuring tape helps to: 1

(a) make the tape heavier

(b) show the time

(c) make the tape bend

(d) hold the tape against the edge of the object so that measuring starts from zero

10. Why are cooking utensils such as pans often coated with a non-stick layer? 1

(a) to make them dissolve **(b)** to make them heavier

(c) to make them transparent **(d)** food does not stick to the surface

11. Temperature of a cup of tea is 80 °C, the room is 30 °C. After a long time, the tea's temperature will be about: 1

(a) 50 °C **(b)** 30 °C **(c)** 80 °C **(d)** 100 °C

12. A farmer in a dry region wants to conserve water for irrigation. A good method is: 1

(a) flooding the field **(b)** drip irrigation
(c) leaving taps open **(d)** cutting trees

13. 3500 m expressed in km is: 1

(a) 0.35 km **(b)** 3.5 km **(c)** 35 km **(d)** 350 km

14. Which of the following pairs includes only soluble substances? 1

(a) sand and chalk **(b)** salt and sugar **(c)** chalk and sugar **(d)** sand and salt

15. Mercury level before measuring should be: 1

(a) at 100 °C **(b)** at 42 °C **(c)** at 40 °C **(d)** below 35 °C

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Convert 3.75 km into metres and centimetres. 2

17. Why do boats made of iron float while iron nails sink? 2

18. A clinical thermometer reading is 39 °C. Interpret it. 2

19. Why does the temperature stay constant at 100 °C when water boils continuously? 2

20. Distinguish between rotation and revolution with an example. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A ribbon 2.4 m long is cut into pieces of 15 cm. Find the number of pieces and the leftover length. 3

22. You are given a mixture of sand, iron filings and sugar. Describe how to separate them using properties. 3

23. A student reads a thermometer as 70 °C for hot water and 25 °C for the room. Describe what will happen to the water temperature in 2 hours. 3

24. Describe the water cycle and name its stages. 3

OR

A pendulum takes 42 s for 21 oscillations. Find its time period and the number of oscillations in 1 minute. 3

25. Give reasons: 3
(a) a doormat is rough;
(b) a tabletop is smooth;
(c) a window is transparent.

OR

Why are the walls of a thermos flask silvered? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain how you would measure the circumference of a cylindrical glass and its diameter, and verify the ratio. 5

OR

Plan an investigation to find the best material for making a thermos flask lid among wood, plastic, steel and cork. 5

27. A student heats water in a beaker and records the temperature every 2 minutes: 30, 38, 46, 54, 62 °C. Find the rise per 2 minutes and the likely temperature after 10 minutes. 5

OR

Describe an experiment to show the effect of surface area, temperature and wind on evaporation. 5

END OF QUESTION PAPER — Total: 50 Marks

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